

Sharp Boolean–Tutte Completion for Finite Random-Cluster Gate Inequalities

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Abstract

The previously proposed Christoffel–Tutte hierarchy applies Cauchy–Schwarz to intersection indicators. It is valid, but it is not the strongest conclusion carried by the same intersection moments. We replace it by an exact finite linear program on gate signatures. The new bound is the largest lower bound valid for every finite event system with the stated moments, dominates the Christoffel/Rayleigh bound at every level, and is strictly stronger in general. Its permutation-invariant quotient is a finite factorial-moment program. At order two this gives the sharp Dawson–Sankoff bound, which strictly improves the second-moment bound and is already exact for two gates. For finite random-cluster measures the same construction has an order-free component-partition expansion. Every dual certificate then gives a coefficientwise nonnegative Tutte-polynomial inequality, valid simultaneously for every cluster weight $q > 0$.

The requested conclusion for all vertex-transitive graphs is not asserted. Taken literally it is false when $p_c = 1$ (for example on $\mathbb{Z}$), and the corrected assertion for all transitive graphs with $p_c < 1$ remains strictly stronger than the results proved here. The Hutchcroft/Kruskal program supplied with the question still requires its stated first-reconvergence or nested anti-congestion estimate. The finite theorem below does not imply that estimate without new quantitative input.

1 Why the Gram hierarchy is not optimal

Let $(\Omega, \mathbb{P})$ be a finite probability space, let I be a nonempty finite set, and let $(E_i)_{i \in I}$ be events. Put

$$D = \bigcup_{i \in I} E_i.$$

Suppose that $X \geq 0$ and $X = 0$ off D . For every nonempty $S \subseteq I$ write

$$J_S = \mathbf{1}_{\cap_{i \in S} E_i}, \quad m_S = \mathbb{E}[X J_S].$$

The Christoffel construction forms the matrix $M_{S,T} = m_{S \cup T}$ and uses

$$\mathbb{E}[X] \geq \frac{(c^\top m)^2}{c^\top M c}. \tag{1.1}$$

This is Cauchy–Schwarz and is correct. It nevertheless discards the most important fact about the indicators J_S : they are monomials of one Boolean signature. Retaining that fact gives an exact linear program.

The loss is already visible for two events. If

$$a = \mathbb{E}[X \mathbf{1}_{E_1}], \quad d = \mathbb{E}[X \mathbf{1}_{E_2}], \quad c = \mathbb{E}[X \mathbf{1}_{E_1 \cap E_2}],$$

then the three numbers used in the degree-one Gram matrix determine the answer exactly:

$$\mathbb{E}[X] = a + d - c. \quad (1.2)$$

By contrast, when $ad > c^2$, the optimized Gram value is

$$\frac{ad(a + d - 2c)}{ad - c^2},$$

and its deficit from (1.2) is

$$\frac{c(a - c)(d - c)}{ad - c^2}. \quad (1.3)$$

Thus a genuine strengthening must use the Boolean constraints rather than another reformulation of the same positive-semidefinite matrix.

2 The sharp signature program

For $\omega \in \Omega$, define its gate signature

$$\sigma(\omega) = \{i \in I : \omega \in E_i\}.$$

For every nonempty $\sigma \subseteq I$, put

$$z_\sigma = \mathbb{E}[X; \sigma(\omega) = \sigma]. \quad (2.1)$$

Then

$$\mathbb{E}[X] = \sum_{\emptyset \neq \sigma \subseteq I} z_\sigma, \quad m_S = \sum_{\sigma \supseteq S} z_\sigma. \quad (2.2)$$

Fix a family $\mathcal{M} \subseteq \mathcal{P}(I) \setminus \{\emptyset\}$ of moments that are to be retained. For a vector $m = (m_S)_{S \in \mathcal{M}}$ define

$$\mathcal{P}_{\mathcal{M}}(m) = \left\{ (u_\sigma)_{\emptyset \neq \sigma \subseteq I} : u_\sigma \geq 0, \sum_{\sigma \supseteq S} u_\sigma = m_S \text{ for every } S \in \mathcal{M} \right\}$$

and

$$\mathcal{B}_{\mathcal{M}}(m) = \min_{u \in \mathcal{P}_{\mathcal{M}}(m)} \sum_{\emptyset \neq \sigma \subseteq I} u_\sigma. \quad (2.3)$$

The minimum exists whenever the data arise from an event system: the actual vector (2.1) is feasible, and intersecting the feasible set with the sublevel set determined by its total mass makes the optimization compact.

Theorem 2.1 (Sharp Boolean moment completion). *For every finite event system and every $X \geq 0$ supported on D ,*

$$\boxed{\mathbb{E}[X] \geq \mathcal{B}_{\mathcal{M}}(m)}. \quad (2.4)$$

Moreover,

$$\boxed{\mathcal{B}_{\mathcal{M}}(m) = \max \left\{ \sum_{S \in \mathcal{M}} c_S m_S : \sum_{\substack{S \in \mathcal{M} \\ S \subseteq \sigma}} c_S \leq 1 \text{ for every } \emptyset \neq \sigma \subseteq I \right\}}. \quad (2.5)$$

The maximum is attained. Among bounds depending only on the listed moments and valid for all finite event systems, (2.4) is best possible.

Proof. The actual signature masses z_σ satisfy (2.2), so they are feasible in (2.3). Their objective is $\mathbb{E}[X]$, proving (2.4).

Let A be the zero-one matrix with rows indexed by $S \in \mathcal{M}$, columns indexed by nonempty $\sigma \subseteq I$, and entries $A_{S,\sigma} = \mathbf{1}_{\{S \subseteq \sigma\}}$. The primal program is

$$\min\{\mathbf{1}^\top u : Au = m, u \geq 0\}.$$

Finite-dimensional linear-programming duality gives

$$\max\{c^\top m : A^\top c \leq \mathbf{1}\},$$

which is exactly (2.5). For completeness, weak duality follows from

$$c^\top m = c^\top Au = (A^\top c)^\top u \leq \mathbf{1}^\top u.$$

For the reverse inequality, let β be the primal minimum and consider the finitely generated closed cone

$$K = \{(Au, \mathbf{1}^\top u + t) : u \geq 0, t \geq 0\}.$$

For every $\varepsilon > 0$, the point $(m, \beta - \varepsilon)$ is outside K . Separating this point from K gives (y, s) with $s > 0$, $A^\top y + s\mathbf{1} \geq 0$, and $y^\top m + s(\beta - \varepsilon) < 0$. The vector $c = -y/s$ is dual feasible and satisfies $c^\top m > \beta - \varepsilon$. Hence the dual supremum is β ; the standard finite-polyhedron attainment argument gives a maximizer.

It remains to justify the optimality claim. Let u be any feasible vector in (2.3). Make one atom ω_σ for every nonempty signature, together with one dummy atom, give all these atoms equal positive probability, put $\omega_\sigma \in E_i$ exactly when $i \in \sigma$, and choose the value of $X(\omega_\sigma)$ so that its expected mass is u_σ ; put $X = 0$ at the dummy atom. This realizes the prescribed moment vector and has $\mathbb{E}[X] = \sum_\sigma u_\sigma$. In particular, an optimizer in (2.3) is realized by a finite event system. No universally valid function of the listed moments can therefore give a larger lower bound. $\square$

Adding moments can only shrink the feasible set in (2.3). Thus the bounds are monotone under enrichment of $\mathcal{M}$. If all nonempty subsets of I are included, Möbius inversion determines every z_σ , and

$$\boxed{\mathcal{B}_{\mathcal{P}(I) \setminus \{\emptyset\}}(m) = \sum_{\emptyset \neq S \subseteq I} (-1)^{|S|+1} m_S = \mathbb{E}[X].} \quad (2.6)$$

This is finite inclusion–exclusion, now identified as the terminal point of an information-optimal hierarchy.

2.1 Using bounded observables

When $0 \leq X \leq 1$, one can retain the gate-signature probabilities

$$h_\sigma = \mathbb{P}(\sigma(\omega) = \sigma).$$

Then $0 \leq z_\sigma \leq h_\sigma$. Define the capped program

$$\mathcal{B}_{\mathcal{M}}^{\text{cap}}(m, h) = \min \left\{ \sum_\sigma u_\sigma : Au = m, 0 \leq u_\sigma \leq h_\sigma \right\}. \quad (2.7)$$

Proposition 2.2 (Capped dual). *If $0 \leq X \leq 1$ and $X = 0$ off D , then*

$$\mathbb{E}[X] \geq \mathcal{B}_{\mathcal{M}}^{\text{cap}}(m, h) \geq \mathcal{B}_{\mathcal{M}}(m), \quad (2.8)$$

and

$$\mathcal{B}_{\mathcal{M}}^{\text{cap}}(m, h) = \max_{c \in \mathbb{R}^{\mathcal{M}}} \left\{ c^{\top} m - \sum_{\sigma \neq \emptyset} h_{\sigma} ((A^{\top} c)_{\sigma} - 1)_+ \right\}. \quad (2.9)$$

This is the best universal lower bound determined by (m, h) .

Proof. The actual z is feasible in (2.7), and the capped feasible set is contained in the uncapped one. For a multiplier c for $Au = m$, minimizing the Lagrangian over the box $0 \leq u \leq h$ gives

$$c^{\top} m + \sum_{\sigma} \min_{0 \leq u_{\sigma} \leq h_{\sigma}} (1 - (A^{\top} c)_{\sigma}) u_{\sigma},$$

which is the expression in (2.9). Finite linear-programming duality gives equality. Conversely, every vector $0 \leq u \leq h$ can be realized on a space with signature masses h_{σ} by setting $X = u_{\sigma}/h_{\sigma}$ on the σ -atom (and setting it to zero when $h_{\sigma} = 0$). This proves optimality. $\square$

3 Comparison with the Christoffel bound

Theorem 3.1 (The Boolean program dominates the Gram program). *Let $\mathcal{F}$ be a family of nonempty subsets of I , and suppose $\mathcal{M}$ contains every set $S \cup T$ with $S, T \in \mathcal{F}$. Form*

$$m_S^{\mathcal{F}} = m_S, \quad M_{S,T}^{\mathcal{F}} = m_{S \cup T}.$$

Then

$$\mathcal{B}_{\mathcal{M}}(m) \geq \sup_{c^{\top} M^{\mathcal{F}} c > 0} \frac{(c^{\top} m^{\mathcal{F}})^2}{c^{\top} M^{\mathcal{F}} c}. \quad (3.1)$$

The inequality can be strict.

Proof. Take any feasible signature vector u in (2.3), and regard u as a finite measure on the nonempty signatures. On this space let $J_S(\sigma) = \mathbf{1}_{\{S \subseteq \sigma\}}$. Its listed first and second moments are exactly $m^{\mathcal{F}}$ and $M^{\mathcal{F}}$. Applying Cauchy–Schwarz under this measure shows

$$\sum_{\sigma} u_{\sigma} \geq \frac{(c^{\top} m^{\mathcal{F}})^2}{c^{\top} M^{\mathcal{F}} c}$$

for every admissible c . Minimize the left side over feasible u and then maximize the right side over c . Strictness follows from the two-event calculation (1.2)–(1.3). $\square$

For the four-cycle data quoted in the preceding draft,

$$a = d = \frac{325}{398}, \quad c = \frac{625}{796}.$$

The Gram value is

$$\frac{8450}{10149} \approx 0.832594,$$

whereas the Boolean program, using exactly the same three moments, gives

$$a + d - c = \frac{675}{796} \approx 0.847990.$$

Thus it recovers the exact answer one full level earlier than the proposed Gram hierarchy.

4 The sharp gate-count hierarchy

The full signature program distinguishes the gates. A much smaller program results if only permutation-invariant moment sums are retained. Let $n = |I|$ and

$$N = \sum_{i \in I} \mathbf{1}_{E_i}.$$

On the support of X , one has $1 \leq N \leq n$. Define the factorial moments

$$S_j = \mathbb{E} \left[X \binom{N}{j} \right] = \sum_{\substack{T \subseteq I \\ |T|=j}} m_T, \quad 1 \leq j \leq n. \quad (4.1)$$

For $1 \leq d \leq n$, put

$$\mathcal{D}_d(S) = \min \left\{ \sum_{k=1}^n u_k : u_k \geq 0, \quad \sum_{k=1}^n \binom{k}{j} u_k = S_j \quad (1 \leq j \leq d) \right\}. \quad (4.2)$$

Theorem 4.1 (Sharp factorial-moment completion). *For every d ,*

$$\boxed{\mathbb{E}[X] \geq \mathcal{D}_d(S)}, \quad (4.3)$$

where

$$\boxed{\mathcal{D}_d(S) = \max \left\{ \sum_{j=1}^d a_j S_j : \sum_{j=1}^d a_j \binom{k}{j} \leq 1 \text{ for } k = 1, \dots, n \right\}}. \quad (4.4)$$

This is the best bound based only on $S_1, \dots, S_d$. The sequence is nondecreasing in d , and

$$\boxed{\mathcal{D}_n(S) = \mathbb{E}[X]}. \quad (4.5)$$

Proof. Put

$$u_k = \mathbb{E}[X; N = k].$$

These nonnegative numbers are feasible in (4.2), their total is $\mathbb{E}[X]$, and (4.4) is the finite LP dual. Every feasible (u_k) is realized by a finite event system supported on signatures of the indicated sizes, so the bound is information-optimal. Enriching the moment list shrinks the primal feasible set. Finally,

$$\sum_{j=1}^n (-1)^{j+1} \binom{k}{j} = 1 \quad (1 \leq k \leq n).$$

Using these coefficients in (4.4) gives $\mathbb{E}[X]$, proving (4.5). $\square$

The order-two program has a closed form. This is the classical sharp Dawson–Sankoff lower bound.

Theorem 4.2 (Sharp order-two formula). *Assume $S_1 > 0$ and put*

$$x = 1 + \frac{2S_2}{S_1}, \quad r = \lfloor x \rfloor. \quad (4.6)$$

Then $1 \leq r \leq n$ and

$$\boxed{\mathcal{D}_2(S) = \frac{2S_1}{r+1} - \frac{2S_2}{r(r+1)}}. \quad (4.7)$$

If $S_1 = 0$, then $X = 0$ almost surely and all quantities vanish. The bound (4.7) dominates the Cauchy–Schwarz estimate

$$\frac{S_1^2}{S_1 + 2S_2}, \quad (4.8)$$

and their difference is

$$\boxed{\frac{S_1(x-r)(r+1-x)}{xr(r+1)} \geq 0}. \quad (4.9)$$

It is strict unless x is an integer.

Proof. For every integer $k \geq 1$ and every positive integer r ,

$$1 - \left(\frac{2k}{r+1} - \frac{2\binom{k}{2}}{r(r+1)} \right) = \frac{(k-r)(k-r-1)}{r(r+1)} \geq 0. \quad (4.10)$$

The last inequality holds because no integer lies strictly between r and $r+1$. Multiplying (4.10) by u_k and summing proves that the right side of (4.7) is a valid dual bound.

To prove optimality, let $\eta = x - r$. Since x is the mean of k under the probability weights ku_k/S_1 , one has $1 \leq x \leq n$ and $0 \leq \eta < 1$, except that $x = n$ simply gives $\eta = 0$ and $r = n$. Define

$$u_r = \frac{S_1(1-\eta)}{r}, \quad u_{r+1} = \frac{S_1\eta}{r+1}, \quad (4.11)$$

with the second term omitted when $r = n$, and set all other u_k to zero. Then

$$\sum ku_k = S_1, \quad \sum \binom{k}{2} u_k = S_2.$$

Its total mass is exactly the right side of (4.7), so the primal and dual values agree. Since $S_1 + 2S_2 = xS_1$, direct subtraction of (4.8) from (4.7) gives (4.9). $\square$

When $n = 2$, write $a = m_{\{1\}}$, $d = m_{\{2\}}$, and $c = m_{\{1,2\}}$. Then $S_1 = a + d$ and $S_2 = c$; Theorem 4.2 gives $a + d - c = \mathbb{E}[X]$. Thus the order-two integer constraint closes exactly the deficit (1.3) left by the continuous Cauchy–Schwarz relaxation.

5 Application to the random-cluster model

Let $G = (V, E)$ be a finite loopless multigraph with edge parameters $p_e \in [0, 1]$ and cluster weight $q > 0$. Its random-cluster law is

$$\phi_{G,p,q}(\omega) = \frac{1}{Z_{G,p,q}} q^{k_G(\omega)} \prod_{e \in \omega} p_e \prod_{e \notin \omega} (1 - p_e). \quad (5.1)$$

No correlation assumption and no restriction on q is needed below.

Fix nonempty vertex sets $R, A \subseteq V$, and let

$$C_R = \bigcup_{r \in R} C_r, \quad D = \{R \leftrightarrow A\}.$$

For each $a \in A$, take the single gate event

$$E_a = \{R \leftrightarrow a\}. \quad (5.2)$$

Then $D = \bigcup_{a \in A} E_a$. In particular, only $|A|$ Boolean gates are required; indexing instead by $R \times A$ is unnecessary.

Let $F : 2^V \rightarrow \mathbb{R}$ be arbitrary and put

$$m_F = \min_{K \subseteq V} F(K), \quad X_F = (F(C_R) - m_F)\mathbf{1}_D. \quad (5.3)$$

For nonempty $S \subseteq A$, define

$$m_S(F) = \phi[F(C_R) - m_F; R \leftrightarrow a \text{ for every } a \in S]. \quad (5.4)$$

Theorem 5.1 (Sharp random-cluster gate completion). *For every $q > 0$, every moment family $\mathcal{M}$, and every real function F ,*

$$\boxed{\phi[F(C_R); D] - m_F \phi(D) \geq \mathcal{B}_{\mathcal{M}}((m_S(F))_{S \in \mathcal{M}}).} \quad (5.5)$$

The capped version of Proposition 2.2 applies whenever $0 \leq F(C_R) - m_F \leq 1$. The factorial-moment bounds of Theorems 4.1 and 4.2 apply after summing (5.4) over sets of equal size. With every nonempty $S \subseteq A$ retained, (5.5) is an equality after finitely many levels.

Proof. Apply Theorem 2.1 to the finite edge-configuration space, the events (5.2), and the non-negative variable (5.3). Equation (5.4) is exactly the corresponding intersection moment. The remaining assertions follow from Proposition 2.2 and Theorems 4.1–4.2. $\square$

For a target vertex b , take $F(K) = \mathbf{1}_{\{b \in K\}}$. Then

$$\alpha_b = \phi(R \leftrightarrow b, R \leftrightarrow A)$$

and

$$p_S(b) = \phi(R \leftrightarrow b, R \leftrightarrow a \text{ for every } a \in S). \quad (5.6)$$

Consequently,

$$\boxed{\alpha_b \geq \mathcal{B}_{\mathcal{M}}((p_S(b))_{S \in \mathcal{M}}).} \quad (5.7)$$

This bound is valid for $q < 1$, $q = 1$, and $q > 1$. For a single source $R = \{o\}$, the moment in (5.4) is gate-readable: on the event in (5.4), $C_a = C_o$ for every $a \in S$.

6 Order-free Tutte certificates

The random-cluster specialization has a further algebraic structure. Use the exact cluster-law reduction: delete edges with $p_e = 0$, contract the components formed by edges with $p_e = 1$, delete loops, and replace a parallel class B by one edge of parameter

$$1 - \prod_{e \in B} (1 - p_e).$$

It is therefore enough to consider a simple graph with $0 < p_e < 1$. Put

$$v_e = \frac{p_e}{1 - p_e}.$$

For a nonempty vertex set B , define

$$\kappa(B) = \sum_{\substack{T \subseteq E(B) \\ (B,T) \text{ connected}}} \prod_{e \in T} v_e, \quad (6.1)$$

with $\kappa(\{v\}) = 1$. For a partition π of V , put

$$w(\pi) = \prod_{B \in \pi} \kappa(B), \quad \mathcal{Z}(q) = \sum_{\pi} q^{|\pi|} w(\pi). \quad (6.2)$$

The open-component partition Π then satisfies

$$\phi(\Pi = \pi) = \frac{q^{|\pi|} w(\pi)}{\mathcal{Z}(q)}. \quad (6.3)$$

Indeed, after the global factor $\prod_e (1 - p_e)$ is removed, a configuration with component partition π is exactly an independent choice of a connected spanning edge set inside every block of π .

For a partition π , let

$$B_R(\pi) = \bigcup \{B \in \pi : B \cap R \neq \emptyset\}, \quad \sigma_R(\pi) = A \cap B_R(\pi).$$

For every nonempty $\sigma \subseteq A$, define the atom polynomial

$$W_{\sigma}^b(q) = \sum_{\substack{\pi: b \in B_R(\pi) \\ \sigma_R(\pi) = \sigma}} q^{|\pi|} w(\pi), \quad (6.4)$$

and the gate-only atom polynomial

$$H_{\sigma}(q) = \sum_{\pi: \sigma_R(\pi) = \sigma} q^{|\pi|} w(\pi). \quad (6.5)$$

All their coefficients are nonnegative, and coefficientwise $0 \leq W_{\sigma}^b \leq H_{\sigma}$. Put

$$L_b(q) = \sum_{\sigma \neq \emptyset} W_{\sigma}^b(q), \quad P_S^b(q) = \sum_{\sigma \supseteq S} W_{\sigma}^b(q). \quad (6.6)$$

Then

$$\alpha_b = \frac{L_b(q)}{\mathcal{Z}(q)}, \quad p_S(b) = \frac{P_S^b(q)}{\mathcal{Z}(q)}.$$

Theorem 6.1 (Coefficientwise Boolean–Tutte certificate). *Suppose $(c_S)_{S \in \mathcal{M}}$ satisfies*

$$a_{\sigma} := \sum_{\substack{S \in \mathcal{M} \\ S \subseteq \sigma}} c_S \leq 1 \quad (\emptyset \neq \sigma \subseteq A). \quad (6.7)$$

Then

$$\boxed{L_b(q) - \sum_{S \in \mathcal{M}} c_S P_S^b(q) = \sum_{\sigma \neq \emptyset} (1 - a_{\sigma}) W_{\sigma}^b(q)}, \quad (6.8)$$

so the polynomial on the left has nonnegative coefficients. In particular, after division by $\mathcal{Z}(q)$, the corresponding random-cluster inequality holds simultaneously for every $q > 0$.

More generally, for arbitrary c put $u_\sigma = (a_\sigma - 1)_+$. Then

$$L_b(q) \geq \sum_{S \in \mathcal{M}} c_S P_S^b(q) - \sum_{\sigma \neq \emptyset} u_\sigma H_\sigma(q), \quad (6.9)$$

and the difference is again coefficientwise nonnegative.

Proof. Expand the right side of (6.8) using (6.6) and collect the coefficient of each atom polynomial W_σ^b . Condition (6.7) makes every multiplier nonnegative.

For (6.9), the contribution of a fixed signature to the difference between the left and right sides is

$$(1 - a_\sigma)W_\sigma^b + u_\sigma H_\sigma.$$

If $a_\sigma \leq 1$, this is nonnegative. If $a_\sigma > 1$, it equals

$$(a_\sigma - 1)(H_\sigma - W_\sigma^b),$$

which is coefficientwise nonnegative. Summing over signatures proves the claim. $\square$

This is stronger than pointwise positive semidefiniteness of the former Christoffel–Tutte matrix in two ways: it is optimal for the chosen Boolean moment data, and every fixed dual witness proves an identity with visibly nonnegative coefficients rather than merely a nonnegative value at one q .

7 What this does not prove on transitive graphs

The statement

$$\theta_G(p_c(G)) = 0 \quad \text{for every infinite vertex-transitive graph } G \quad (7.1)$$

is false without a qualification. For nearest-neighbour bond percolation on $\mathbb{Z}$, one has $p_c = 1$ and $\theta(1) = 1$. The intended statement is the conjecture obtained by adding $p_c(G) < 1$.

The Hutchcroft/Kruskal notes supplied with the question do not prove that corrected statement. Their final displayed nested anti-congestion limit is an additional theorem-sized hypothesis. Calling it a consequence of transitivity would be circular unless it is proved.

There is a second logical separation. Theorems 2.1 and 5.1 optimally convert known finite intersection moments into a union estimate. They do not bound those intersection moments. In the replica construction, the missing first-reconvergence estimate is exactly a bound on such overlaps. An information-optimal union inequality cannot create an overlap estimate that has not been supplied.

The order-two theorem does sharpen the bookkeeping in the proposed low-congestion/high-congestion dichotomy. If N counts any finite family of replica reconvergence events and

$$S_1 = \mathbb{E}N, \quad S_2 = \mathbb{E}\binom{N}{2},$$

then Theorem 4.2 is the exact smallest possible value of $\mathbb{P}(N > 0)$ compatible with these two factorial moments. Equivalently, for every positive integer r ,

$$\mathbb{P}(N > 0) \geq \frac{2S_1}{r+1} - \frac{2S_2}{r(r+1)}. \quad (7.2)$$

Thus a large first-reconvergence charge either forces a large union or a quantitatively large pair-overlap charge. What remains absent is a theorem turning the latter charge, uniformly over scales, into percolation below p . The bundle-canopy example in the supplied notes shows that unimodularity, finite mean degree, and critical survival alone do not give that conversion.

Accordingly, the strongest rigorous conclusion is:

- the finite random-cluster hierarchy is completely strengthened by the sharp Boolean and capped programs above;
- the permutation-invariant order-two relaxation has the closed sharp formula (4.7), and the full finite hierarchy is exact;
- the component-partition expansion promotes every dual witness to a coefficientwise Tutte-polynomial inequality for all $q > 0$;
- no unconditional extension to all transitive graphs follows until the stated anti-congestion or an equivalent robustness theorem is proved.